

A Machine-Checked Anytime-Valid Confidence Sequence by the Method of Mixtures

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Abstract

We give a machine-checked anytime-valid confidence sequence, formalized in Lean 4 against Mathlib and reducing to only the three standard classical axioms (`propext`, `Classical.choice`, `Quot.sound`), with no `sorry`, `admit`, or added axioms. The construction is the method of mixtures: a continuous mixture over a compactly supported uniform tilt prior of sub-Gamma exponential supermartingales, with a countable-time Ville inequality. For a predictable-increment process bounded by b , conditionally centered with respect to the past, with conditional second moment at most σ^2 , the mixture process $\bar{M}_n$ exceeds $1/\delta$ at some $n \geq 1$ with probability at most δ , for every $\delta \in (0, 1)$ and admissible window $0 < \lambda_0 < \lambda_1 < 3/b$. The measurability and product-integrability package the argument needs is discharged inside the proof from the increment model alone, so the headline carries no free joint- measurability or integrability hypotheses, and the kernel rechecks the guarantee on demand from the proof term and a pinned toolchain.

Keywords: anytime-valid inference, confidence sequences, method of mixtures, Ville’s inequality, formal verification, Lean 4

1. Anytime-valid guarantees as a verification target

A confidence sequence is a family of interval estimates that all hold at once uniformly over time, with probability at least $1 - \delta$. This time-uniform validity is the backbone of safe, anytime-valid inference (Howard et al., 2021; Ramdas et al., 2023) and the sequential testing that conformal and distribution-free prediction lean on. The standard route builds a nonnegative supermartingale, optimizes a free tilt by mixing against a prior (Robbins, 1970), and applies Ville’s inequality (Howard et al., 2020; Shafer and Vovk, 2019).

The argument is short to state and delicate to discharge. The mixing step exchanges a conditional expectation with a prior integral, needing a Fubini hypothesis and the defining set-integral property of conditional expectation, and the supermartingale property needs joint measurability and product-integrability over (prior $\times$ sample) space. Informal treatments elide these as routine; a machine-checked statement cannot, so in the Lean 4 (de Moura and Ullrich, 2021) / Mathlib (The mathlib Community, 2020) formalization reported here every side condition is an explicit hypothesis or it is proved.

* A large-language-model coding assistant was used for proof engineering; all statements are kernel-checked.

2. Construction and the discharged headline

Fix a real increment process $(X_k)_k$ on a filtered probability space $(\Omega, (\mathcal{F}_n)_n, \mu)$ and write $S_n = \sum_{i < n} X_i$. We use the conditional sub-Gamma increment model, with the standard martingale-difference indexing: the increment X_k is revealed one step ahead, so X_k is $\mathcal{F}_{k+1}$ -measurable (predictable-increment adaptedness), is almost surely bounded ($|X_k| \leq b$), is conditionally centered with respect to the past ($\mu[X_k \mid \mathcal{F}_k] = 0$), and is conditionally variance-controlled ($\mu[X_k^2 \mid \mathcal{F}_k] \leq \sigma^2$). The one-step-ahead indexing is what keeps the centering non-vacuous: requiring X_k to be $\mathcal{F}_k$ -measurable instead would force $\mu[X_k \mid \mathcal{F}_k] = X_k$, collapsing the centered process to zero. The running sum S_n stays $\mathcal{F}_n$ -measurable because it uses only $X_0, \dots, X_{n-1}$. For a fixed tilt λ the sub-Gamma exponential process is $M_\lambda(n) = \exp(\lambda S_n - n\psi(\lambda))$, with cumulant proxy $\psi(\lambda) = \sigma^2 \lambda^2 / (2(1 - b\lambda/3))$; under the model each M_λ is a nonnegative supermartingale. The mixture averages these against a continuous prior on the tilt, $\bar{M}_n = \int M_\lambda(n) d\rho(\lambda)$ with $\rho = \text{Unif}[\lambda_0, \lambda_1]$ on a compact admissible window. It inherits the supermartingale property by exchanging the conditional expectation with the prior integral, which we prove by testing on $\mathcal{F}_n$ -measurable sets; this avoids any joint measurability requirement in the tilt parameter.

Theorem 1 (`mixture_confidence_sequence_uniformPrior`) *Let $\delta \in (0, 1)$, $b > 0$, $\sigma^2 \geq 0$, and $0 < \lambda_0 < \lambda_1 < 3/b$. For a predictable-increment process satisfying the conditional sub-Gamma model above, the mixture process $\bar{M}_n$ satisfies*

$$\mu\left(\left\{\omega : \exists n \geq 1, \frac{1}{\delta} \leq \bar{M}_n(\omega)\right\}\right) \leq \delta.$$

That is, the mixture crosses $1/\delta$ with probability at most δ over all n , and inverting the threshold yields a time-uniform confidence sequence for the increment mean. The proof invokes seven supporting results directly and twenty-three transitively: the mixture is a nonnegative supermartingale (prior-integral exchange plus the fixed-tilt one-step bound), and the countable-time Ville inequality gives the crossing bound from $\mathbb{E}[\bar{M}_0] = 1$. The measurability and product-integrability obligations are not assumed; for the uniform prior they are discharged internally from the pointwise bound $M_\lambda(n) \leq \exp(\lambda_1 n b)$, so the headline needs no joint-measurability or product-integrability hypothesis over the prior-sample space; it assumes only per-increment measurability and integrability, the increment model, and the parameter constraints. A `#print axioms` audit (Lean 4 v4.32.0, Mathlib 81a5d25, FormalSLT e0b1c79) reduces the headline and every supporting lemma to the three standard classical axioms (`propext`, `Classical.choice`, `Quot.sound`), with no `sorry`, `admit`, or custom axioms; a worked instance with a genuine nonzero increment (a ± 1 Rademacher sign on $\Omega = \{0, 1\}$, revealed at step one, with $\delta = 1/2$, $b = \sigma^2 = 1$, $[\lambda_0, \lambda_1] = [1/2, 1]$) satisfies all hypotheses at once and audits to the same three axioms, so the guarantee is non-vacuous on a nondegenerate process. The result sits in a Lean library of finite-sample statistical learning theory whose claims audit to the same axioms.

3. Positioning and scope

This is a machine-checked anytime-valid confidence sequence, complementary to the toolkit COPA serves: time-uniform supermartingale bounds of this kind are the sequential back-

bone underneath betting-style and distribution-free coverage (Waudby-Smith and Ramdas, 2024; Vovk et al., 2005). Prior learning-theory formalizations target a different lane, empirical-process and PAC bounds (Sonoda et al., 2025; Karayel and Tan, 2023; Bagnall and Stewart, 2019; Tassarotti et al., 2021); we are not aware of a prior machine-checked anytime-valid confidence sequence by the method of mixtures, and claim only that result.

Three boundary conditions delimit the result: the conditional sub-Gamma increment model (bounded, centered, variance-controlled), not a heavy-tailed process; a uniform prior on a compact window $[\lambda_0, \lambda_1] \subset (0, 3/b)$, not an unbounded or optimized- λ mixture; and a countable-time, not continuous-time, guarantee. The next step is to widen the prior family, the only dependency being a new pointwise process bound.

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